TOPS Technologies

Interview Questions — Hardware and Components Basics Understanding of Hardware and its components

Total Questions: 29

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1. What is the function of the motherboard in a computer system?

> **Hint:** Think about how different hardware components communicate with each other.

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2. Name two main differences between HDD and SSD storage devices.

> **Hint:** Consider speed and internal technology.

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3. Why is RAM important for a computer’s performance?

> **Hint:** Focus on temporary data storage and multitasking.

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4. What does a CPU do in a computer system?

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5. If you had to label the following components on a PC diagram—CPU, RAM, GPU, Storage—where would you typically find each on the motherboard?

> **Hint:** Think about the physical slots and sockets.

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6. Explain the importance of compatibility between a CPU and a motherboard.

> **Hint:** Focus on sockets and chipsets.

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7. What is the role of the chipset on a motherboard?

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8. How would you identify the PCIe slot on a motherboard?

> **Hint:** Look for the longest expansion slot, often used for GPUs.

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9. A motherboard has an LGA1200 socket. What does this tell you about CPU compatibility?

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10. Describe the steps you would take to safely remove and reinstall a RAM module.

> **Hint:** Think about power, static discharge, and the RAM clips.

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11. What is the main difference between DDR3, DDR4, and DDR5 RAM?

> **Hint:** Consider speed, power consumption, and physical notch location.

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12. How would you tell if a drive is an NVMe SSD just by looking at it and its connection?

> **Hint:** Focus on the connector type and slot position.

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13. When would you choose an SSD over an HDD for a new computer build?

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14. What is the primary function of a GPU in a computer?

> **Hint:** Think about graphics and display output.

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15. Explain the difference between air cooling and liquid cooling in PCs.

> **Hint:** Consider how heat is transferred away from components.

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16. How would you identify a GPU inside a desktop PC case?

> **Hint:** Look for its location and connection to the motherboard.

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17. A user wants to upgrade their graphics for gaming. What should you check on the motherboard before installing a new GPU?

> **Hint:** Check for available PCIe slots and power connectors.

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18. Describe one advantage and one disadvantage of liquid cooling compared to air cooling.

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19. Walk me through the process of assembling a desktop PC from scratch. What are the key steps you would document?

> **Hint:** Start from motherboard installation and end with power-up.

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20. During disassembly, why is it important to document and label each step?

> **Hint:** Think about troubleshooting and reassembly.

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21. If a PC does not power on after assembly, what are the first three things you would check?

> **Hint:** Consider power connections, RAM seating, and motherboard standoffs.

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22. How would you safely handle and install a CPU to avoid damaging it?

> **Hint:** Remember static discharge and pin alignment.

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23. You have a motherboard, CPU, RAM, GPU, SSD, and HDD. How do you determine the correct order to install these components during assembly?

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24. What could happen if you forget to connect the CPU fan during assembly?

> **Hint:** Think about overheating and system protection features.

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25. Explain how you would use an AI assistant like ChatGPT to help you troubleshoot a PC that won’t boot, and what steps you’d take to double-check the advice you receive.

> **Hint:** Focus on using AI for step-by-step guidance but verifying with official documentation.

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26. How could you use AI image recognition tools to help identify unknown hardware components inside a PC case, and what are some limitations of relying solely on AI for this task?

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27. Integrative: If you upgrade to DDR5 RAM and a new GPU, what motherboard features must you check to ensure compatibility with both components?

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28. Integrative: How does the choice of storage type (HDD vs. NVMe SSD) affect the assembly process and the placement of components on the motherboard?

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29. Integrative: During PC assembly, how do cooling system choices impact the installation and positioning of the CPU, GPU, and RAM?
